



Generative Adversarial Network for Imputation of Road Network Traffic State Data

Dongwei Xu^{1,2} , Zefeng Yu^{1,2}, Tian Tian³, and Yanfang Yang⁴ 

¹ Institute of Cyberspace Security, Zhejiang University of Technology,
Hangzhou 310023, China

² College of Information Engineering, Zhejiang University of Technology,
Hangzhou 310023, China

³ Hangzhou Zhongao Technology Company Limited, Hangzhou 310000, China

⁴ China Academy of Transportation Sciences, Beijing 100029, China
yangyf@motcats.ac.cn

Abstract. The loss of traffic state data is a common problem in intelligent transportation system. To improve the imputation accuracy and robustness of road traffic data, a novel generative adversarial network for the imputation of road network traffic data is proposed in this paper, i.e., GAE-GAN-LSTM. The spatiotemporal characteristics were extracted using a improved graph auto-encoder (GAE), followed by a generative adversarial network (GAN) to generate the complete spatiotemporal characteristics on the basis of the missing features. The internal structure of the generator was a long short-term memory network (LSTM), and the internal structure of the discriminator was a fully connected neural network (FCN). Finally, the traffic state data could be recovered by the decoder of GAE. The experimental results revealed that the performance of the proposed method was better than that of the other methods at any data loss ratio considered. The main innovations of the proposed method include two aspects. One, an improved GAE for the imputation of road network data was presented by redefining the loss function of GAE, which could effectively extract the potential spatiotemporal features of a road network. Two, the GAN was used to generate the spatiotemporal characteristics of the traffic state data by using the strong data generation ability of GAN.

Keywords: Data imputation · Graph auto-encoder · Generative adversarial network · Road traffic state data

1 Introduction

With the increase in urbanization, urban road traffic problems are becoming worse. Traffic management and control engineering need complete and precise road traffic state data, which mainly includes vehicle speed and traffic flow.

These traffic state data are not only temporally correlated, but also spatially correlated on adjacent lanes; therefore, road traffic state data are critical for intelligent transport systems [1]. Because of a road detector or transmission failure and other reasons [2], the road traffic state data may be missing, which may have an adverse effect on an intelligent transportation system. Therefore, data imputation has become a research hotspot of transportation [3].

Over the years, there have been many methods of traffic state data imputation, which contains three types of methods, that is, prediction-based data imputation method, interpolation-based data imputation method, and statistical-based data imputation method [4].

The prediction-based data imputation method predicts the missing data from the relationship among the historical data to realize the imputation of traffic state data, such as the autoregressive integrated moving average (ARIMA) model [5], the feed-forward neural network method [6, 7], and support vector regression (SVR) [8], as well as some of their variants [9]. Although some improved methods can also improve the precision of imputation of traffic state data, the precision of data imputation decreases when there is continuous missing data or a large proportion of missing data.

The interpolation-based data imputation method filling the loss data depends on the historical data, such as, moving average method, exponential smoothing method, historical average method, local least squares (LLS) [10], and k-nearest neighbor method (KNN) [11, 12] and its variant [13]. The interpolation-based data imputation can provide better imputation results by relying on the assumption that the adjacent traffic state data are similar. However, the adjacent traffic state data do not always meet this assumption, as the traffic state data change in the time series. Therefore, there are some shortcomings of this method, which result in unsatisfactory data imputation results.

The statistical-based data imputation method presets the probability distribution model according to the data and then fills in the loss data according to the distribution of the maximum probability of the model, such as Markov chain Monte Carlo (MCMC)-based interpolation method [14, 15], probabilistic principal component analysis-based interpolation method (PPCA) [16], functional principal component analysis-based interpolation method (FPCA) [17], and kernel probabilistic principal component analysis (KPPCA) [18]. The statistics-based data imputation method needs the data to satisfy the probabilistic distribution; therefore, the method has a shortcoming that the data do not always meet the probability distribution model.

In addition, deep learning has also been used in intelligent transport system and data imputation, which has a stronger ability of data extraction and data generation. The deep learning methods are also gradually applied to imputation of traffic state data, such as denoising stacked autoencoder method (DSAE) [19] and importance-weighted autoencoder (MIWAE) [20]. GAN [21], proposed in 2014, is a framework of deep learning. GAN consists of two parts: one is generator and the other is discriminator. The generator can generate data, and the discriminator can judge the authenticity of the generated data. GAN is

widely used in image super-resolution [22], image dataset synthesis [23] and text generation. Recently, GAN gradually applied to data imputation [24, 25]. GAN has the problems of unstable training and mode collapse, but the above methods can not solve these problems. Thus, in this paper, we use Wasserstein-GAN (WGAN) to address this challenge. In the field of Intelligent Transport Systems, GAN has made a considerable contribution to traffic state data imputation [25] and traffic flow prediction [26, 27]. Moreover, LSTM has extensive applications in traffic speed prediction [28]. However, these methods in processing of traffic state data can not accurately extract the graph structure of traffic state data and spatiotemporal characteristics. Graph auto-encoder (GAE), serving as a variant of Graph convolution Network (GCN), takes GCN as the encoder, and thus, the graph structure of traffic state data can be extracted effectively.

For these problems and challenges, to improve the precision and the robustness of traffic state data imputation, a novel generative adversarial network for the imputation of road network traffic state data has been proposed, i.e., GAE-GAN-LSTM. Firstly, GAE has been used to extract the spatiotemporal characteristics of the missing dataset, followed by GAN to generate the spatiotemporal characteristics of the complete dataset according to the characteristics of the missing dataset. In particular, the internal structure of the generator is LSTM, and the internal structure of the discriminator is a fully connected neural network. Finally, the traffic state data imputation can be realized by the decoder of GAE.

Our innovations are as follows:

- (1) By redefining the loss function of graph autoencoder, we developed an improved graph autoencoder suitable for road network data repair, which can effectively extract the potential spatiotemporal characteristics of the road network.
- (2) Because of the strong data generation ability of the generated adversarial network, we used this network to generate the spatiotemporal characteristics of the traffic state data.
- (3) Compared with other models, the proposed model can enhance the precision of data repair, and maintain good performance and strong robustness in a high loss ratio.

2 Methodology

Figure 1 shows the framework of our method. First, a GAE model was applied to extract the spatiotemporal characteristics of the incomplete traffic state data. Second, the complete traffic spatiotemporal characteristics were generated according to the spatiotemporal characteristics of the incomplete data by GAN. The generator and the discriminator were the LSTM and the fully connected neural network, respectively. The generator is to generate the Spatiotemporal characteristics of the complete data according to the spatiotemporal characteristics of the incomplete data. The function of the discriminator is to judge the authenticity of the Spatiotemporal characteristics generated by generator,

so as to ensure that the spatiotemporal features generated are similar to the spatiotemporal features of the complete data. Finally, the complete data were generated using the decoder of GAE, and the imputation of the traffic state data was realized.

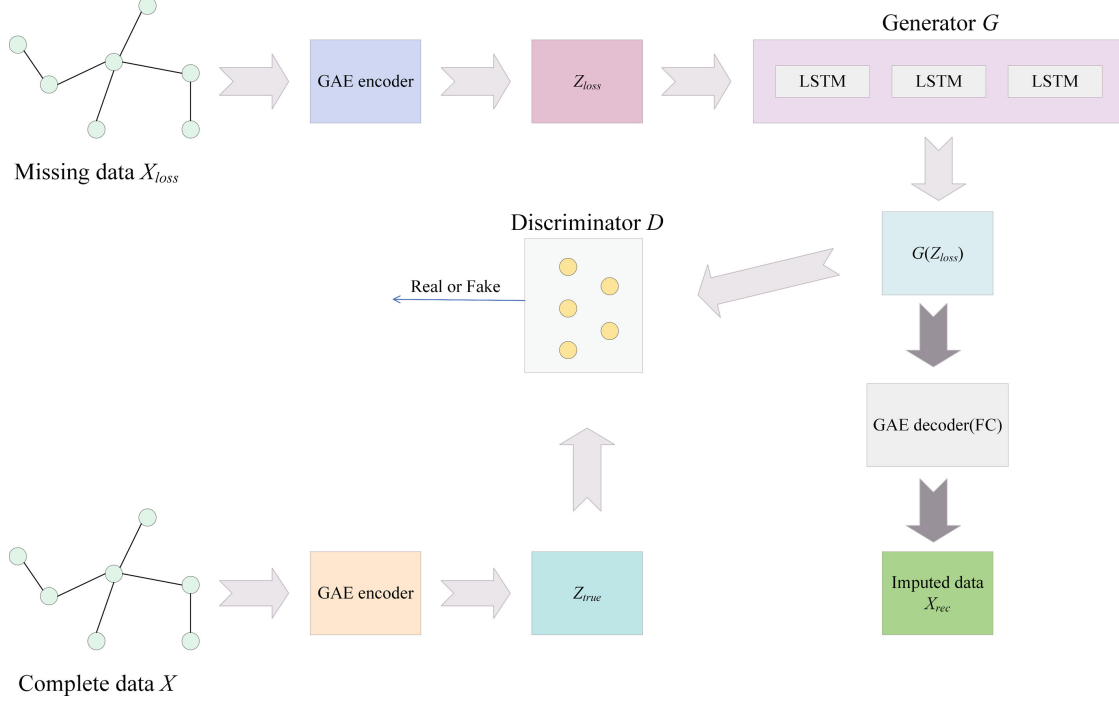


Fig. 1. Proposed generative adversarial network for imputation of road network traffic state data

2.1 Data Preprocessing

Observing that the spatial relationship of different detectors and the temporal relationship between the data of each detector, we modeled the road network as $\mathcal{G} = (\mathcal{V}, \mathcal{E})$. $\mathcal{V} = \{v_i\}_{i=1}^N$ denotes N detectors and $\mathcal{E} = \{e_{ij}\}_{i,j=1}^n$ denotes the connection relationship between two corresponding links. $|\mathcal{V}| = N$ denotes the number of nodes. If v_i and v_j are spatially adjacent, $e_{ij} = 1$; else, $e_{ij} = 0$. The traffic state data can be described as follows:

$$X = \begin{bmatrix} x_{11} & x_{21} & \dots & x_{N1} \\ x_{12} & x_{22} & \dots & x_{N2} \\ \vdots & \vdots & \ddots & \vdots \\ x_{1M} & x_{2M} & \dots & x_{NM} \end{bmatrix} \quad (1)$$

where N denotes the number of traffic detectors and M represents the number of traffic state data items when the time interval of sampling is t . x_{ij} represents the j -th traffic state data of the detector i . By normalization processing, we

translated the elements of X between 0 and 1. In addition, the missing mask matrix Q of the traffic state data with a certain loss ratio could be described as follows:

$$Q = \begin{bmatrix} q_{11} & q_{21} & \dots & q_{N1} \\ q_{12} & q_{22} & \dots & q_{N2} \\ \vdots & \vdots & \dots & \vdots \\ q_{1M} & q_{2M} & \dots & q_{NM} \end{bmatrix} \quad (2)$$

where q_{it} indicates whether x_{it} is missing. If so, $q_{ij} = 1$; otherwise, $q_{ij} = 0$. The traffic state data including the missing data can be denoted as $X_{loss} = X_{true} * Q$, and $*$ denotes the multiplication operation of the corresponding elements of the two matrices. The training and testing datasets were divided under a certain proportion of p . The aim of the model was to repair the road network data by extracting the spatiotemporal characteristics of the road network using the missing road network data X_{loss} . The imputation data was denoted as X_{rec} .

2.2 The Feature Abstraction of Road Network Based on GAE

Construction of GAE. GAE is one of the unsupervised learning methods, including an encoder and a decoder. Compared to the autoencoder, the encoder of GAE is GCN and the decoder is a graph decoder. In this study, the GAE was improved and the loss function of GAE was redefined. Figure 2 shows the framework of improved GAE. The improved GAE exhibited better performance for extracting the spatiotemporal characteristics of the road network by encoding and decoding the input data. The expressions of GAE could be described as follows:

$$Z = GCN(X, A) \quad (3)$$

$$\hat{X} = \sigma_2(W_{D1}(\sigma_1(W_{D0}Z + B_{D0})) + B_{D1}) \quad (4)$$

$$\hat{A} = \sigma_3(ZZ^T) \quad (5)$$

where traffic state data X represents the input matrix of the encoder; A is the adjacent matrix, which is equivalent to $\mathcal{E}$. Z represents the spatiotemporal characteristics extracted after encoding. $\hat{X}$ represents the new traffic state output data matrix of the decoder. $\sigma_1, \sigma_2, \sigma_3$ represent the activation function. W_{D0} and W_{D1} are the weighted matrices of the decoder. B_{D0} and B_{D1} represent the bias matrix. $\hat{A}$ is the new adjacent matrix of the decoder's output. Z^T is the transposed matrix of Z . By minimizing the difference between $\hat{A}$ and the real adjacency matrix A , the potential spatiotemporal characteristics Z in the traffic state data is extracted. The extracted spatiotemporal characteristics Z can be decoded by a fully connected neural network and the new traffic state data $\hat{X}$ is obtained. The encoder can also be described as follows:

$$\tilde{A} = D^{-\frac{1}{2}}AD^{-\frac{1}{2}} \quad (6)$$

$$GCN(A, X) = \tilde{A}\sigma_4(\tilde{A}XW_{G0})W_{G1} \quad (7)$$

where W_{G0} and W_{G1} represent the weighted matrix of the encoder and σ_4 represents the activation function. D is the degree matrix.

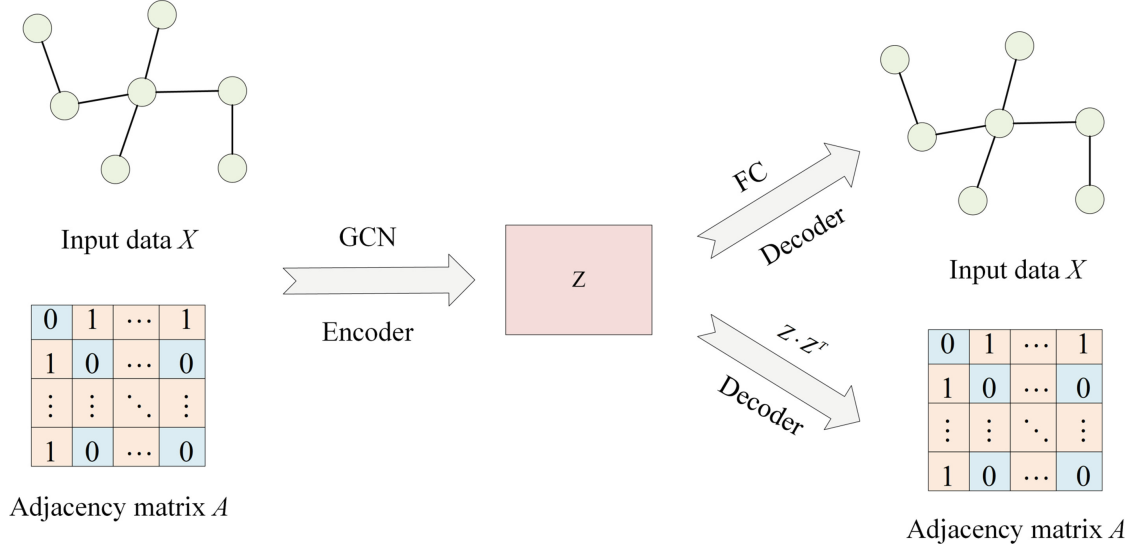


Fig. 2. The framework of the improved GAE

Loss Function of GAE. In this paper, we reconstruct the decoder of GAE to meet our work, which not only retains the decoding of adjacency matrix, but also adds the decoding of traffic state data, so that GAE can not only give play to the advantages of extracting graph features, but also be used to decode traffic state data. Therefore, the loss function of GAE should be composed of two parts, one is the loss function suitable for traffic state data decoding, L_D , and the other is the loss function suitable for adjacency matrix decoding, L_A . They are connected by a super parameter α to obtain the final loss function of gae, L_G :

$$L_G = L_D + \alpha L_A \quad (8)$$

where L_D was constructed by mean square error:

$$L_D = \frac{1}{p} \sum_{i=0}^p (\hat{X} - X)^2 \quad (9)$$

where p denotes the number of data items. In this paper, the loss function of weighted cross entropy with logits was used to construct L_A :

$$q = \frac{N^2 - \sum_{i=1}^N \sum_{j=1}^N a_{ij}}{\sum_{i=1}^N \sum_{j=1}^N a_{ij}} \quad (10)$$

$$l = (1 + (q - 1) * \hat{A}) \quad (11)$$

$$r = \frac{N^2}{N^2 - 2 \sum_{i=1}^N \sum_{j=1}^N a_{ij}} \quad (12)$$

$$g = (1 - A) * \hat{A} + l * (\log(1 + \exp(-\text{abs}(\hat{A}))) + \max(\hat{A}, 0)) \quad (13)$$

$$L_A = r \frac{g}{N^2} \quad (14)$$

where a_{ij} denotes an element of the original matrix in x row j column, q denotes the weight of cross entropy, and l, r, g denote the intermediate transition variables in the calculation process.

2.3 The Design of Generative Adversarial Network for Spatio-Temporal Feature Based on LSTM

Construction of GAN-LSTM. In this section, GAN-LSTM generates the spatiotemporal characteristics of the complete data based on the missing spatiotemporal characteristics extracted by using the GAE method. GAN-LSTM contains generator G and discriminator D . The purpose of G is to generate data that D supposes to be true. While the purpose of D is to distinguish the real data as true and the data generated by the generator as false. Through constant adversarial training, G can generate data that are more similar to the real data.

The internal structure of generator G is LSTM, followed by a fully connected network to obtain the output. The expression of LSTM can be described as follows:

$$f_t = \sigma(W_f[h_{t-1}, x_t] + b_f) \quad (15)$$

$$i_t = \sigma(W_i[h_{t-1}, x_t] + b_i) \quad (16)$$

$$\tilde{C}_t = \tanh(W_c[h_{t-1}, x_t] + b_C) \quad (17)$$

$$C_t = f_t * C_{t-1} + i_t * \tilde{C}_t \quad (18)$$

$$o_t = \sigma(W_o[h_{t-1}, x_t] + b_o) \quad (19)$$

$$h_t = o_t * \tanh(C_t) \quad (20)$$

where W_f, W_i, W_c, W_o and b_f, b_i, b_C, b_o represent the weight matrix and bias matrix, respectively. f_t, i_t and o_t represent the output of the forget gate, input gate, and the output gate, respectively. $\tilde{C}_t$ and C_t represent the temporary and the new states of the memory cell, respectively. σ represent the Sigmoid function. $\tanh(\cdot)$ can be described as follows:

$$\tanh(\varepsilon) = \frac{e^\varepsilon - e^{-\varepsilon}}{e^\varepsilon + e^{-\varepsilon}} \quad (21)$$

The generator G is implemented using LSTM, followed by a fully connected network to obtain G_Z :

$$G_Z = \sigma(W_G h_t + b_G) \quad (22)$$

where W_G and b_G denote the weighted and bias matrix, respectively. The internal structure of discriminator D is an n -layer fully connected network whose first layer is as follows:

$$Y_R^{(1)} = W_d^{(1)} Z + B_1 \quad (23)$$

$$Y_F^{(1)} = W_d^{(1)} G_Z + B_1 \quad (24)$$

The expressions of the i -layer fully connected network are as follows:

$$Y_R^{(i)} = W_d^{(i)} Y_R^{(i-1)} + B_i \quad (25)$$

$$Y_F^{(i)} = W_d^{(i)} Y_F^{(i-1)} + B_i \quad (26)$$

Finally, $Y_F^{(i)}$ and $Y_R^{(i)}$ represent the results of the i -layer fully connected network of generator G_Z and the hidden layer feature matrix of GAE, respectively.

Loss Function of Generative Adversarial Network. In this section, Wasserstein-GAN is used to replace GAN. Different from the traditional GAN, WGAN uses the Wasserstein distance rather than the JS divergence to estimate the similarity of the data distribution, because WGAN can deal with the vanishing gradient problem in the training process and ensure the diversity of the generated samples. The loss functions of G and D can be listed as follows:

$$D_L = \frac{1}{k} \sum_{i=1}^k y_F^{(n)(i)} - \frac{1}{k} \sum_{i=1}^k y_R^{(n)(i)} \quad (27)$$

$$G_L = \frac{1}{k} \sum_{i=1}^k y_F^{(n)(i)} - \frac{1}{p} \sum_{i=1}^p (G_i - Z_i) \quad (28)$$

where k and p denote the output number of discriminator D and generator G , respectively. $y_F^{(n)(i)}$, $y_R^{(n)(i)}$, G_i and Z_i represent the i -th element of $Y_F^{(n)}$, $Y_R^{(n)}$, G_Z and Z respectively. D_L and G_L represent the loss function of discriminator D and generator G , respectively.

2.4 The Imputation of Road Network Data Based on GAE-GAN

To better apply GAE to road network data imputation, data decoding was added to the decoder of GAE. Fully connected layers were used for the data decoding, and the traffic state data imputation was realized through GAE decoding based on G_Z . The decoding process can be described as follows:

$$X_{rec} = \sigma(W_{D1}(ReLU(W_{D0}G_Z + B_{D0})) + B_{D1}) \quad (29)$$

Finally, X_{rec} was de-normalized to obtain the traffic states data imputation.

3 Experiment

3.1 Experimental Design

The dataset of the Seattle expressway network, which include traffic speed, was used for our experiment. The traffic state data of 23 road detectors were selected from January 1, 2017 to March 1, 2017, and the data sampling interval was 5 min.

In the experiment, the data missing patterns were categorized as missing completely at random (MCAR) or MCAR in space (MCARS). In addition, the missing ratio ranged from 10% to 90%. Figure 3 shows the complete dataset and the data heat map under the MCAR and MCARS data missing modes with different data missing ratios. The X-axis denotes the time on January 1 and the Y-axis denotes the detectors. The left and right subfigures of Figs. 3(b and c) denote the missing patterns of MCAR and MCARS, respectively. Finally, the proposed model was compared with the other baselines under the two data missing patterns.

3.2 Parameter Setting and Model Index

To test the precision of the proposed model, some baselines were compared with the proposed model. First, to prove the effectiveness of GAE's extraction features, the proposed model was compared with GAN-LSTM. In the GAN-LSTM model, the hidden layer of LSTM in generator was 1, and its units was 64. The hidden layer of FCN of discriminator was 3, and its units was [32, 64, 128]. Second, to verify the effectiveness of the generator LSTM, the structure of the generator was revised to FCN, i.e., GAE-GAN-BP. In the GAE-GAN-BP model, D and G each had 3 hidden layers, and their units were [32, 64, 128]. Finally, some typical baselines were compared with the proposed method, such as KNN, DSAE, and LSTM. Both BP and DSAE had 3 hidden layers, and their units were [32, 64, 128]. The value of k in KNN was 5. The LSTM had one hidden layer, and the number of hidden layer units was 64. In the structure of GAE-GAN-LSTM, the units of encoder was [32, 16] and the units of decoder was 32. Other parameters are the same as GAN-LSTM. The α value was 0.0001. σ_1 and σ_4 were the ReLU functions. σ_2 and σ_3 were the Sigmoid functions.

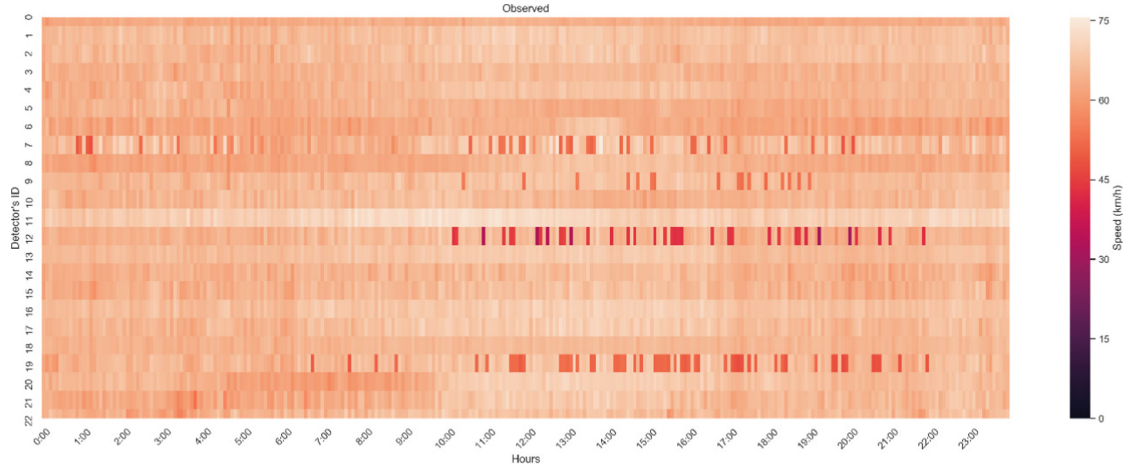
To compare models, three typical evaluation indices, namely root mean square error (RMSE), mean absolute error (MAE), and mean absolute percentage error (MAPE), were adopted in this paper, which are shown as follows:

$$\text{RMSE} = \left[\frac{1}{n} \sum_{i=1}^n (y_i - \tilde{y}_i)^2 \right]^{\frac{1}{2}} \quad (30)$$

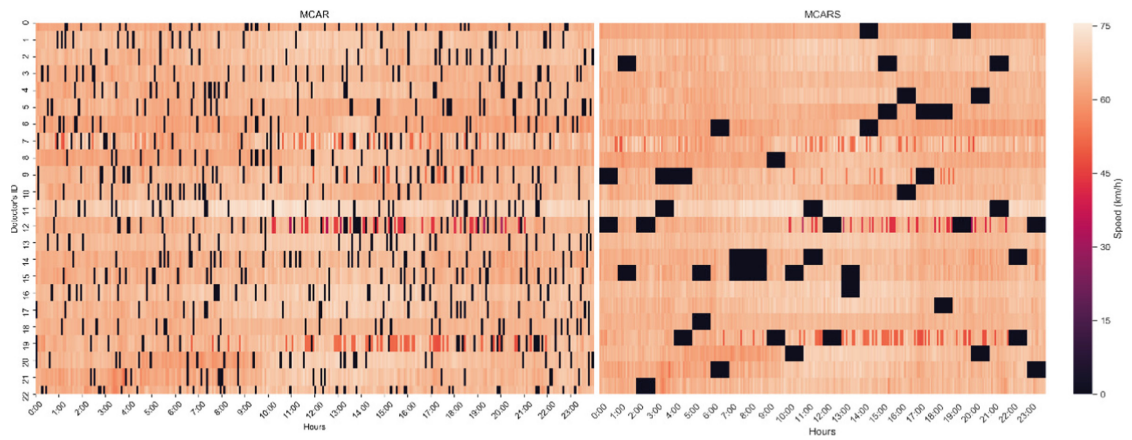
$$\text{MAE} = \frac{1}{n} \sum_{i=1}^n |y_i - \tilde{y}_i| \quad (31)$$

$$\text{MAPE} = \frac{1}{n} \sum_{i=1}^n \left| \frac{y_i - \tilde{y}_i}{y_i} \right| \quad (32)$$

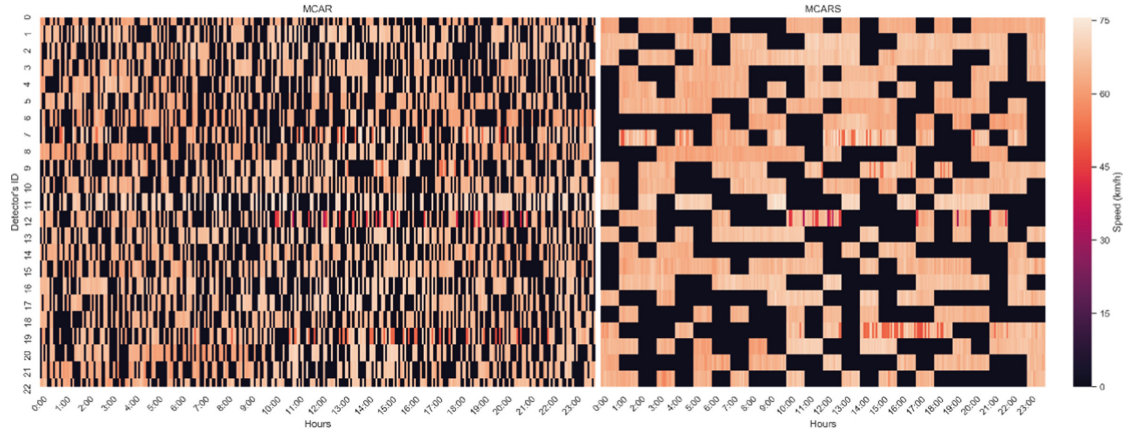
where y_i and $\tilde{y}_i$ denote the true and the imputed values in the missing locations, respectively.



(a) Complete dataset



(b) Speed data of MCAR and MCARS under 10% missing ratio



(c) Speed data of MCAR and MCARS under 50% missing ratio

Fig. 3. Speed data heat map of Seattle expressway network dataset on January 1, 2015

4 Results and Discussion

4.1 Effectiveness of GAE

Compared with the traditional GAE model, the data decoder was merged into the graph decoder of GAE to reconstruct the loss function, which made GAE further extract the spatiotemporal characteristics of the traffic state data. To verify the effectiveness of GAE, the index results of GAE-GAN-LSTM were compared with those of GAN-LSTM. The results of GAE-GAN-LSTM and GAN-LSTM under the MCAR and MCARS modes are shown in Fig. 4.

Figure 4(a) shows that under the MCAR mode, when the data loss ratio was less than 70%, GAE-GAN-LSTM model had a lower error than the GAN-LSTM model, and the error did not significantly increase with an increase in the data loss ratio. When the ratio was higher than 70%, the error increased significantly, but it remained in the low error range and was lower than that of the GAN-LSTM model.

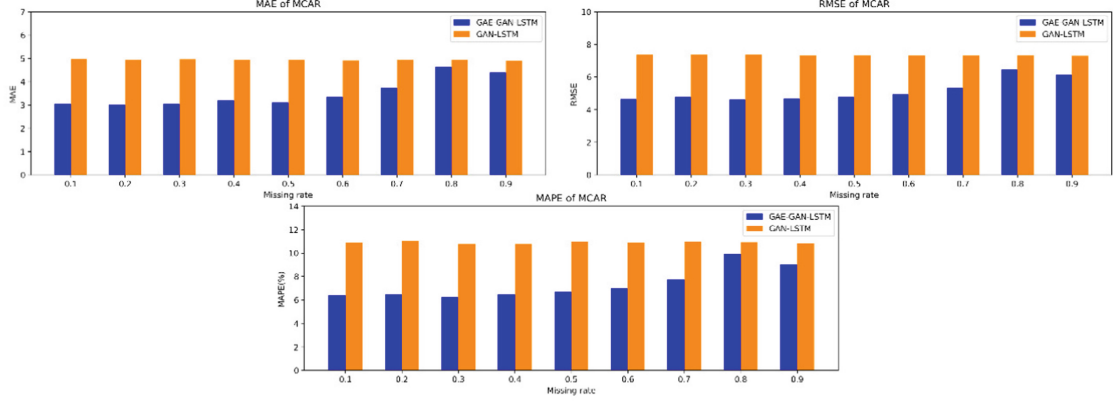
Figure 4(b) shows that under the MCARS mode, the index results of GAE-GAN-LSTM model were similar to those of the MCAR mode. When the data loss ratio was less than 70%, the errors of GAE-GAN-LSTM model were lower than those of GAN-LSTM model. When the data loss was greater than 70%, the errors of GAE-GAN-LSTM were close to those of the GAN-LSTM model. In particular, when the data loss ratio was 90%, the errors of the proposed model were higher than those of GAN-LSTM, but most of the data loss ratio conditions were still satisfied.

From the results shown in Figs. 4(a) and 4(b), the following conclusions could be drawn. The proposed model showed lower error and stronger stability under the MCAR model, which was inseparable from GAE's ability to extract the graph features and the spatiotemporal characteristics of the data. Under the MCARS mode, when the data loss ratio was less than 70%, the model accuracy was higher. When the data loss was higher than 70%, the model precision began to decline, which might be associated with the internal generator of LSTM. Under the MCARS mode, the higher data loss ratio would lead to a continuous data loss phenomenon in the time dimension, which reduced the feature extraction ability of LSTM. Note that the proposed model showed strong stability irrespective of whether the data loss ratio was high or low, which showed the effectiveness of the GAE to extract the spatiotemporal characteristics of the road network data.

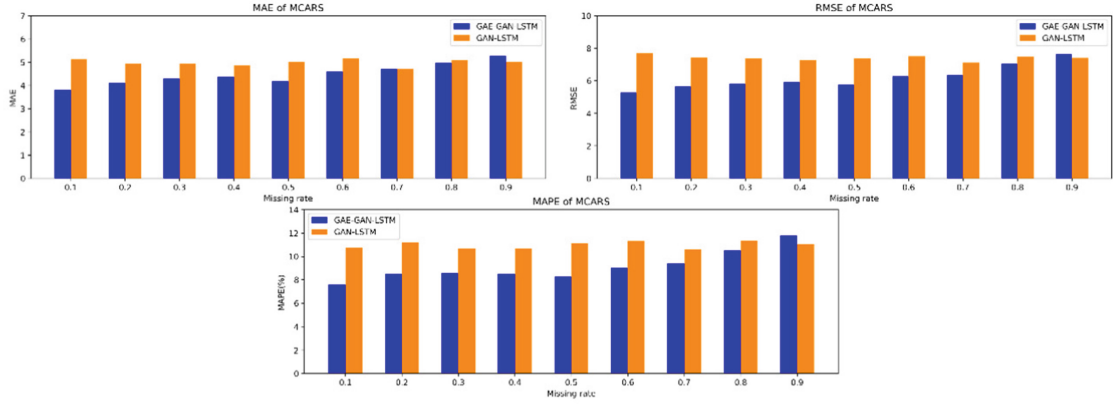
4.2 Effectiveness of Internal Structure of Generator (LSTM)

The internal structure of the generator in the proposed model was LSTM. In this paragraph, the internal structure of the generator was revised to a fully connected neural network for comparison, that is, GAE-GAN-BP. The index results of GAE-GAN-LSTM and GAE-GAN-BP under the MCAR and MCARS modes are shown in Fig. 5.

Figures 5(a) and 4(b) show that the accuracy of the GAE-GAN-LSTM was considerably higher than that of GAE-GAN-BP under the two data loss modes



(a) Histogram results of GAE-GAN-LSTM and GAN-LSTM under the MCAR mode



(b) Histogram results of GAE-GAN-LSTM and GAN-LSTM under the MCARS mode

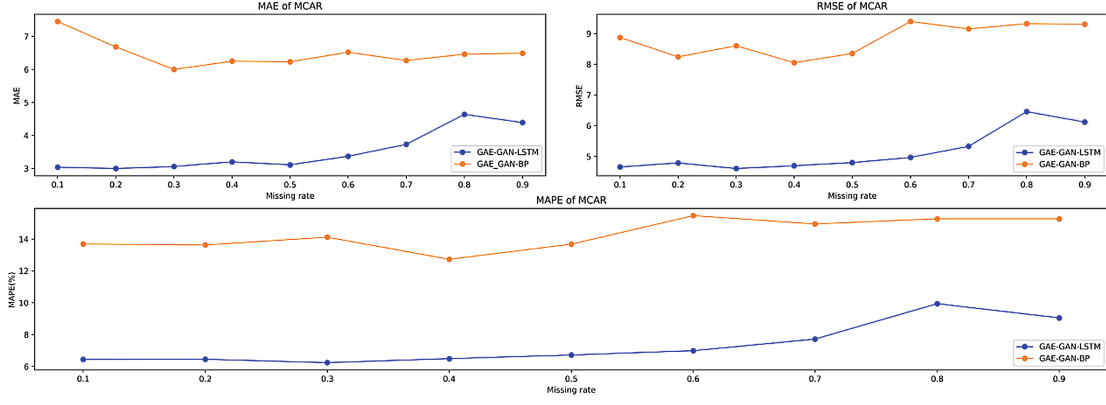
Fig. 4. Histogram results of GAE-GAN-LSTM and GAN-LSTM under different data loss mode

and all the data loss ratios, which was related to the characteristics of the LSTM and BP models. LSTM inherited most of the characteristics of the RNN model and was very suitable for feature extraction in the time series. Therefore, LSTM was suitable as the generator to generate the characteristics. Compared with LSTM, BP was not very sensitive to the characteristics in the time series. Moreover, BP exhibited good performance in the training process but poor performance in the testing process, which was related to the weak ability of generalization. Based on the above analysis, as the internal structure of the generator, LSTM could effectively enhance the precision of the imputation.

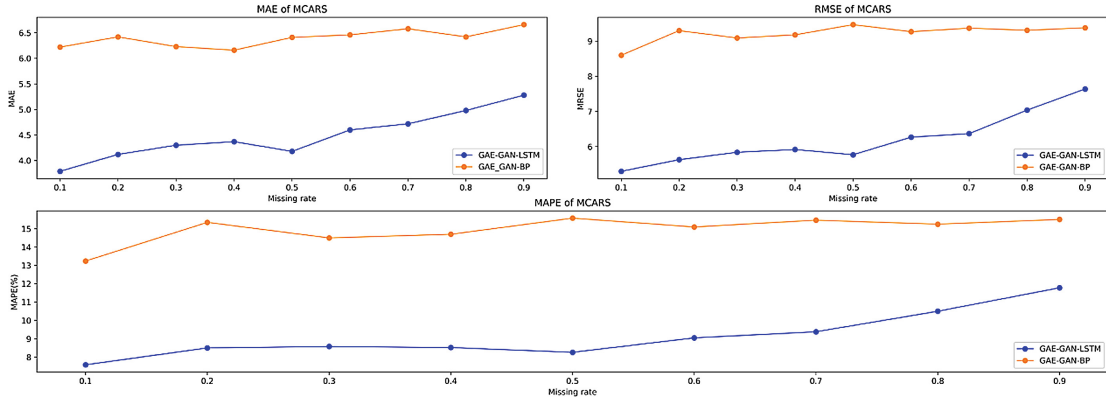
4.3 Comparison with Other Models

Some baselines were compared with the proposed methods under the MCAR and MCARS modes. The loss ratio of the baselines was 10%, 50%, and 90%; Tables 1 and 2 show the corresponding index.

In Table 1, the errors increased with the loss ratio increased. At the low loss ratio of 10%, the proposed models, both BP and LSTM, had a high accuracy.



(a) Results of GAE-GAN-LSTM and GAE-GAN-BP under the MCAR mode



(b) Results of GAE-GAN-LSTM and GAE-GAN-BP under the MCARS mode

Fig. 5. Results of GAE-GAN-LSTM and GAE-GAN-BP under different data loss modes

However, when the loss ratio was increased to 90%, the accuracy of BP declined rapidly. However, the accuracy of the proposed model and LSTM did not decline very quickly and maintained a certain stability, which demonstrated that LSTM had an advantage in the feature extraction in the time-series data. Moreover, as GAE could extract the graph characteristics, the proposed model had better stability than LSTM. However, both the KNN and the DSAE models demonstrated poor accuracy and stability, which might be attributed to the fact that the KNN model required the daily data to have a relative similarity. Moreover, DSAE's generalization ability was not strong.

In Table 2, under the MCARS mode, the errors of most of the models increased, which could be attributed to the characteristics of the MCARS and MCAR modes. Figure 3(b) and 3(c) show that compared with the MCAR mode, under the MCARS mode, the large-area continuous data loss phenomenon was more likely to occur. However, the missing data under the MCAR mode were relatively discrete, which increased the difficulty of feature extraction for the models listed in Table 2, reducing the precision of the models. As for stability, with the loss ratio increased, the error trend under the MCARS mode was

similar to the error trend under the MCAR mode, and the proposed model still had the best effect among the baselines.

Table 1. Index results under MCAR mode

Model	Loss rate								
	10%			50%			90%		
	MAE	RMSE	MAPE (%)	MAE	RMSE	MAPE (%)	MAE	RMSE	MAPE (%)
GAE-GAN-LSTM	3.04	4.66	6.44	3.11	4.8	6.72	4.39	6.12	9.05
BP	3.43	4.96	6.9	4.78	7.03	10.03	5.79	8.84	12.42
KNN	5.6	7.95	11.38	6.22	9.13	12.77	6.32	9.49	13.7
DSAE	5.95	7.22	10.61	6.16	8.23	13.27	6.52	9.16	15.08
LSTM	3.47	5.11	7.45	3.92	5.86	8.68	4.9	7.28	10.71

Table 2. Index results under MCARS mode

Model	Loss rate								
	10%			50%			90%		
	MAE	RMSE	MAPE (%)	MAE	RMSE	MAPE (%)	MAE	RMSE	MAPE (%)
GAE-GAN-LSTM	3.79	5.3	7.58	4.18	5.77	8.26	5.28	7.64	11.78
BP	5.12	7.1	9.71	5.7	8.34	13.04	5.72	8.27	12.83
KNN	7.39	10.46	14.87	6.48	9.86	13.83	6.57	9.86	14.01
DSAE	6.92	9.60	16.44	6.70	9.49	15.81	6.58	9.31	15.27
LSTM	4.71	7.10	10.9	4.88	7.16	10.63	5.96	8.39	12.01

5 Conclusions

Aiming to improve the low precision of road traffic state data imputation, a novel generative adversarial network for the imputation of road network traffic state data was proposed. First, considering that the traffic state data had certain graph structure characteristics, we extracted the spatiotemporal characteristics of the incomplete traffic state data by GAE. Second, owing to the powerful data generation ability of GAN, GAN was applied to generate the features characteristics of the complete data. The internal structure of the generator adopted LSTM, which improved the ability of generating spatiotemporal characteristics. Finally, the data decoding was completed based on the spatiotemporal characteristics generated by the improved decoder of GAE, and the road network traffic state data were repaired. From the analysis, we drew the following conclusions:

- 1) The decoder of GAE was improved to make GAE suitable for the decoding of road network data. Moreover, the GAE could enhance the performance of the model.

- 2) LSTM was used in the generator part of GAN. Compared with the internal structure of the fully connected neural network, LSTM had a better effect on the data imputation and improved the imputation accuracy of the model.
- 3) Compared with the traditional data imputation method, GAE-GAN-LSTM had a higher imputation accuracy. Moreover, the model maintained high imputation precision with high loss ratios. The stability of the proposed model was incomparable to that of the traditional models.
- 4) Because of the strong stability, the proposed model could better satisfy the requirements of a road traffic network with large-scale data missing, which makes a considerable contribution to road traffic state data imputation.

In the future, other datasets will be considered for verification, and the parameters will be modified to improve the repair accuracy. Meanwhile, some parameters will be simplified to make the model more concise and further improve the practicability of the model.

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